

Decision-Relevant Preference Elicitation under a Population Prior

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Abstract

Robots that learn from human preference comparisons ask people which short behaviors look better in order to infer a reward function. Much of the uncertainty about that reward never changes what the robot should actually do, so limited human patience is spent resolving differences with no behavioral consequence. We select trajectory comparisons by expected reduction in policy regret under a population prior over previous users, rather than by parameter information gain, and evaluate on a tabular gridworld with exact optimal policies and exact regret using simulated mixture-population users. A population particle prior substantially reduces final policy regret relative to a uniform particle prior under the same VOI acquisition (C-mech-2; E-prior), and a misspecified prior hurts (C-mech-3; E-prior-shift). Relative to BALD, one-step VOI is not yet a supported win on final regret (C-main-2; E3-power / E3-power-120 / E-decouple-power); VOI remains only fragiley better than random (C-main-1), with a large remaining gap to a true-regret oracle (C-main-3). Changing the acting rule (map / sample / softminimax) does not close that gap (C-mech-4; E-act). These results support using a population prior for decision-relevant elicitation while clarifying that beating parameter-uncertainty acquisition remains open.

Contents

1	Introduction	3
1.1	Contributions	3
2	Related Work	3
2.1	Decision-relevant elicitation	3
2.2	Active preference-based reward learning	4
2.3	Population and pluralistic preference models	4
2.4	Active personalization elsewhere	4
3	Problem Formulation	4
3.1	Reward model	4
3.2	Response model	5
3.3	Population setting	5
3.4	Objective	5
3.5	Oracle	5
4	Method	5
4.1	Problem Setup	5
4.2	Approach	5
4.3	Algorithm	5
4.4	Complexity / Practical Notes	6

5	Experiments	6
5.1	Setup	6
5.2	Main Results	6
5.3	Ablations	6
5.4	Failure Modes	6
6	Discussion	6
6.1	What the results imply	6
6.2	Limitations	6
6.3	Broader impact	7
7	Conclusion	7
A	Additional Experimental Details	8
B	Extended Results	8

1 Introduction

Robots learn reward models from human comparisons of trajectory segments. Because each comparison costs scarce human attention, query selection—which pair to ask next—is the practical budget, not compute. The dominant practice maximizes information about reward parameters [Sadigh et al., 2017, Bıyık et al., 2019].

That target is misaligned with how a robot acts. Information-gain and BALD criteria [Houlsby et al., 2011] shrink uncertainty over parameters; the robot then deploys a *policy* induced by its reward estimate. Prior work already argued that optimizing parameter uncertainty can be wasteful: Model-Free Preference Elicitation [Martin et al., 2024] moved the elicitation target from parameters to reward equivalence classes and recommendation-quality value of information. Independently, Regan and Boutilier [2009] developed regret-based reward elicitation for MDPs, selecting queries to reduce decision-relevant uncertainty about near-optimal policies. We instantiate that decision-theoretic goal for preference-based robotics: score queries by expected reduction in *policy regret* under a *population particle prior* over previous users, rather than searching unrestricted reward space. On our domain, nearby unit rewards can leave the optimal policy unchanged in a local neighborhood at low feature dimension; that local stability collapses as dimension grows—motivating a regret target, not a claim that irrelevance expands with d .

We study binary segment comparisons under Bradley–Terry likelihoods and linear state rewards. For a fixed policy, value is linear in the weight vector, so batch policy regret is exact and cheap. The loop is select (VOI on regret), update the posterior, then decide (act). We evaluate against random sampling, BALD, and a true-regret oracle; we do not claim a new elicitation paradigm, only this instantiation.

1.1 Contributions

1. Formalize preference-query acquisition as expected policy-regret reduction under a population particle prior, using linearity of policy value to compute exact batch regret for fixed policies.
2. Pre-register a paired comparison of random, BALD, particle VOI, and a true-regret oracle under exact policy regret, reporting interval estimates and the remaining oracle gap rather than a bake-off win.
3. Report diagnostics that bound interpretation where evidence exists: a particle-coverage diagnostic for residual regret (no calibrated threshold claimed here), policy-equivalence measurement versus dimension, and consistency-identification negatives (factored consistency latents and difficulty regimes).

2 Related Work

2.1 Decision-relevant elicitation

Regan and Boutilier [2009] develop regret-based reward elicitation for MDPs, selecting queries to reduce decision-relevant uncertainty about near-optimal policies rather than to identify a unique reward; only distinctions that change the optimal (or near-optimal) policy are worth eliciting. Their setting uses bound-style queries over reward structure, and related work extends approximate regret-based elicitation in MDPs [Alizadeh et al., 2015]. Model-Free Preference Elicitation [Martin et al., 2024] likewise moves the elicitation target off raw parameters toward reward equivalence classes and recommendation-quality expected value of information, but in recommender settings without an MDP policy. The decision-relevant / regret-based *goal* itself is not novel for this paper. We instantiate that goal for preference-based robotics: human trajectory-segment comparisons under Bradley–Terry likelihoods rather than bound queries; a learned population prior over previous users rather than searching unrestricted reward space; and a robotics setting with trajectories.

2.2 Active preference-based reward learning

Active preference-based reward learning chooses which trajectory comparison to ask next. Classical criteria shrink uncertainty over reward parameters: volume removal over reward weights [Sadigh et al., 2017], batch volume removal [Biyik and Sadigh, 2018], and information gain that favors answerable questions [Biyik et al., 2019]. Later methods use ensemble disagreement [Hejna and Sadigh, 2023], continuous query synthesis [Zhou and Biyik, 2026], and effort-aware selective querying [Muraleedharan and H, 2025]. Closest within PbRL, Ellis et al. [2024] acquire information about reward *equivalence classes* for a *single user*; our target is instead the regret of the *executed policy* under a *population prior*. Outside robotics, EPIG [Smith et al., 2023] is the general active-learning version of the claim that parameter information can be the wrong target, measuring expected *predictive* mutual information between the query label and future predictions—not policy regret or Regan-style value of information. We keep the segment-comparison interface of active PbRL, but score queries by expected reduction in policy regret under a population prior rather than by parameter volume, information gain, single-user equivalence acquisition, or predictive mutual information.

2.3 Population and pluralistic preference models

A parallel line models preference diversity as structure rather than noise. As shown by MaxMin-RLHF (Theorem 1) [Chakraborty et al., 2024], a single average reward can fail under diverse preferences—the “average user” is nobody. Distributional preference learning accounts for hidden context [Siththaranjan et al., 2024]; pluralistic and prototype-based models personalize reward predictors [Chen et al., 2025]; and clustering methods learn representative rewards from diverse preferences [Kim et al., 2026]. Closest architecturally, variational preference learning encodes annotations into a user latent with a latent-conditioned reward [Poddar et al., 2024], but uses fixed survey queries and leaves active selection as future work. These methods primarily use population structure to *represent* or align to users. We use the population to *choose queries*, not only to model users.

2.4 Active personalization elsewhere

Outside robot policy learning, comparison-based active preference learning personalizes multi-dimensional models in language settings [Oh et al., 2025], and active utility-based pairwise sampling improves personalized recommendations [Boroomand and Wright, 2025]. Neither setting has a downstream policy or a regret target. Across these neighbors, the cell we claim is an instantiation—policy-regret scoring of human trajectory comparisons under a population prior—not a new elicitation paradigm.

3 Problem Formulation

3.1 Reward model

A user is characterized by preference weights $w \in \mathbb{R}^d$ and consistency $b > 0$. States carry features $\phi(s) \in \mathbb{R}^d$; the linear reward is $r(s) = w \cdot \phi(s)$. Under Bradley–Terry, answer probabilities depend on the product $b\|w\|$ only, so only that scale is identifiable: we pin $\|w\| = 1$ and put all scale into b . A trajectory segment is summarized by discounted feature counts

$$f = \sum_t \gamma^t \phi(s_t). \quad (1)$$

Our tabular elicitation experiments use single-state segments (segment horizon $H=1$), so $f = \phi(s)$ there; we do not treat those runs as multi-step discounted occupancy counts.

3.2 Response model

Queries are binary comparisons of segments A and B . With $\|w\| = 1$,

$$P(A \succ B) = \sigma(b w \cdot (f_A - f_B)), \quad (2)$$

where σ is the logistic sigmoid.¹

3.3 Population setting

Previous users $u_1, \dots, u_N$ each provide a small number of comparisons, inducing a *population prior* over (w, b) . A new user arrives with unknown (w, b) . The robot asks T preference queries, then acts by deploying a policy on the MDP. We maintain a particle belief drawn from that prior; acquisition and the select–update–decide loop are deferred to section 4.

3.4 Objective

Let ρ denote the robot’s belief (or the decision induced by it), and let $\pi(\rho)$ be the policy executed after deciding from ρ . Under the true reward w , policy regret is

$$\text{Reg}(\rho) = V_w^* - V_w^{\pi(\rho)}, \quad (3)$$

the gap between optimal value and the value of the executed policy (cf. regret-based elicitation for MDPs [Regan and Boutilier, 2009]; we do not claim the criterion as new). Because the value of a fixed policy is linear in w for tabular linear rewards, this regret is exact and cheap to evaluate in batch. We take *low-budget* mean regret after $t \leq 3$ queries as the **primary**, pre-registered evaluation target; final-query regret is **secondary**. No empiric comparisons are reported in this section.

3.5 Oracle

As an evaluation ceiling we include a true-regret *oracle*: an acquisition that scores candidates using ground-truth regret under the known (w, b) . It is never a baseline to beat; it appears in every elicitation table (section 5) to bound the remaining gap under the same candidate pool and decision rule.

4 Method

4.1 Problem Setup

A user has unit preference weights $w \in \mathbb{R}^d$ ($\|w\| = 1$) and consistency $b > 0$. Trajectory segments are summarized by features f ; a query is a pair (or K -tuple) of segments. Answers follow Bradley–Terry / Plackett–Luce. The robot acts with a policy π on an MDP; regret is $V_w^* - V_w^\pi$.

4.2 Approach

We maintain a belief β as weights over a *fixed* particle set drawn from a population prior over previous users. The myopic value of a query q is the expected reduction in posterior policy loss

$$\text{score}(q) = \text{Loss}(\beta) - \sum_y P_\beta(y \mid q) \text{Loss}(\beta \mid q, y).$$

Unlike BALD, Loss is defined in terms of policy values, not parameter entropy.

4.3 Algorithm

Implementation: `plr.algorithm.POPVOI` (select $\rightarrow$ update $\rightarrow$ decide).

¹Full rankings follow Plackett–Luce; Bradley–Terry is the $K=2$ special case. Main experiments use binary comparisons.

Algorithm 1 POP-VOI (population-prior regret VOI elicitation)

Require: MDP $\mathcal{M}$, population particles $\{(w_p, b_p)\}_{p=1}^P$, candidate queries $\mathcal{Q}$, budget T

- 1: Initialize weights $\beta(w_p) \leftarrow 1/P$
 - 2: **for** $t = 1, \dots, T$ **do**
 - 3: $q_t \leftarrow \arg \max_{q \in \mathcal{Q}} \text{score}(q; \beta)$
 - 4: Observe ranking y_t for q_t
 - 5: $\beta(w_p) \propto \beta(w_p) P(y_t \mid q_t, w_p, b_p)$
 - 6: **end for**
 - 7: **return** π^* for $\hat{w} = \text{normalize}(\mathbb{E}_\beta[w])$
-

Table 1: Placeholder main result table.

Method	Metric A $\uparrow$	Metric B $\downarrow$
Baseline	–	–
Ours	–	–

4.4 Complexity / Practical Notes

Particles and their optimal policies are cached once; only weights change online. Candidate pools stay modest ($|\mathcal{Q}| \sim 10\text{--}50$); binary queries use 2! answer marginalization.

5 Experiments

5.1 Setup

TODO: Datasets / environments, baselines, metrics, seeds, hardware, hyperparameter selection rule (fixed a priori vs. tuned on held-out).

5.2 Main Results

TODO: Primary table/figure answering the success criterion in `PROBLEM.md`.

5.3 Ablations

TODO: Which components matter? Negative controls?

5.4 Failure Modes

TODO: Where the method breaks. Required for honesty, not optional.

6 Discussion

6.1 What the results imply

TODO: Interpret, do not restate tables.

6.2 Limitations

TODO: Scope, data, compute, external validity. Be specific.

6.3 Broader impact

TODO: Dual use, fairness, safety, environmental cost — only where relevant.

7 Conclusion

TODO: Restate the gap closed, the strongest empirical/theoretical claim, and one concrete next step. Keep under half a page.

Acknowledgments

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A Additional Experimental Details

TODO: Hyperparameters, compute budget, full seed tables, prompts, licenses.

B Extended Results

TODO: Extra figures/tables that clutter the main text.